Why

The edge of a wildfire is the most dangerous place for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

> **But a robot near a fire must know where the fire *will* be, not just where it is now.**

How we predict the fire

Real fire history
~246,000 NASA satellite fire detections from 30 major wildfires worldwide

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Weather
wind, humidity, temperature, rain

Terrain
elevation and slope

Land cover
forest, shrub, grass, urban

Fire shape
nearby burning cells, fire size

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Machine learning: XGBoost
27 inputs per 50 m cell. Trained on 24 fires, tested on 6 fires it never saw.

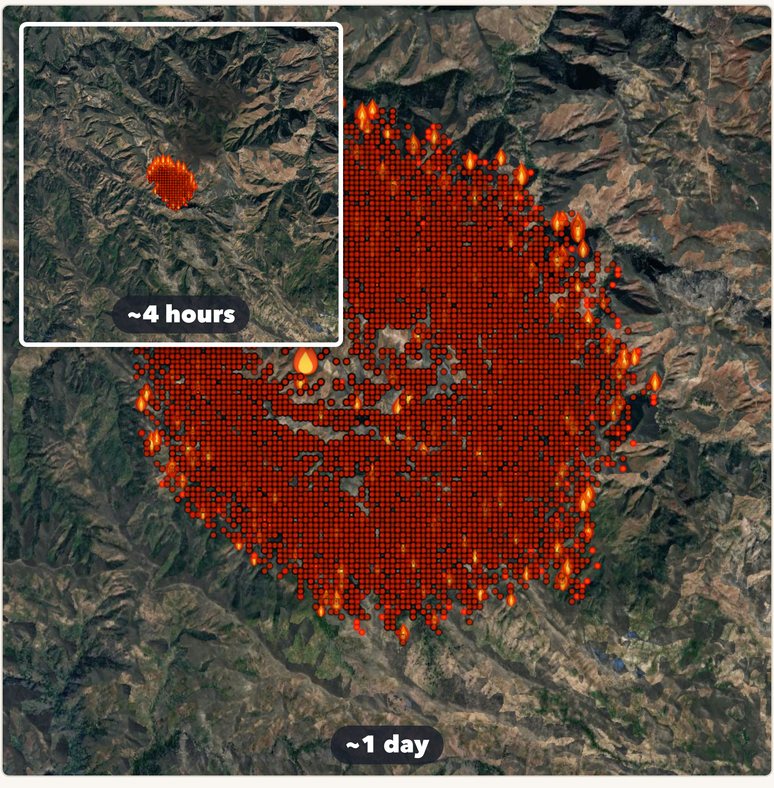
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Physics: Rothermel spread equation
fuel × wind × slope × dryness. The ML model adjusts the rate.

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Output
the chance each 50 m cell catches fire, every time step

The simulator



Simulated fire spreading over real 3D terrain in the Big Sur area of Monterey County, California (summer weather). Runs in a web browser on Google 3D map tiles.